

DAY 02 — Authorized Local Target Lab & Nmap Network Service Assessment

Organization: Trios Cyber

Assessment: Day 02 Assessment

Date: 02 September 2026

Assessment Type: Authorized Local Network Enumeration

Scanner: Kali Linux

Target: Metasploitable

Target IP: 192.168.56.101

Scanner IP: 192.168.56.102

Network: 192.168.56.0/24

Interface: eth0

Virtualization Platform: Oracle VirtualBox

1. Executive Summary

This assessment was conducted as part of the Day 02 cybersecurity internship activity at Trios Cyber.

The objective was to create and validate an authorized local target lab, verify network connectivity between the Kali Linux assessment machine and a Metasploitable target, identify the target IP address, perform basic Nmap host and TCP port scanning, and document the services and software versions discovered.

The assessment was performed entirely within a controlled VirtualBox lab environment using the private 192.168.56.0/24 network.

Kali Linux was configured with the IP address 192.168.56.102/24, while the Metasploitable target was identified at 192.168.56.101/24.

Connectivity was successfully verified using ICMP and neighbour discovery. Nmap subsequently confirmed that the target host was active and identified 23 open TCP ports in the default TCP scan. Service and version detection identified services including FTP, SSH, Telnet, SMTP, DNS, HTTP, SMB/Samba, NFS, MySQL, PostgreSQL, VNC, IRC, Java RMI, and Apache Tomcat.

No exploitation or unauthorized access was performed during this assessment. The activity was limited to network discovery, port enumeration, service identification, and documentation.

2. Assessment Objectives

The primary objectives of the Day 02 assessment were:

1. Create an authorized local target lab using Metasploitable.
 2. Verify connectivity between Kali Linux and the target.
 3. Identify and document the target IP address.
 4. Perform Nmap host discovery.
 5. Perform a basic TCP port scan.
 6. Identify services and software versions using Nmap service detection.
 7. Document discovered services and security-relevant observations.
 8. Preserve command output and screenshots as technical evidence.
-

3. Authorization and Scope

All activities described in this report were conducted against a deliberately vulnerable Metasploitable virtual machine under the user's control.

Authorized Scope

Parameter	Value
Assessment Machine	Kali Linux
Scanner IP	192.168.56.102
Target Machine	Metasploitable
Target IP	192.168.56.101
Network	192.168.56.0/24
Interface	eth0
Environment	Oracle VirtualBox
Assessment Type	Local network enumeration

Scope Restriction

The assessment was restricted to the local VirtualBox laboratory environment.

No external, public, third-party, or unauthorized systems were scanned.

No exploitation, credential attacks, persistence, privilege escalation, or destructive actions were performed.

4. Lab Environment

The assessment consisted of two virtual machines connected through a private VirtualBox network.

Assessment Machine

Operating System: Kali Linux

Interface: eth0

IPv4 Address: 192.168.56.102/24

Target Machine

Operating System: Metasploitable

Interface: eth0

IPv4 Address: 192.168.56.101/24

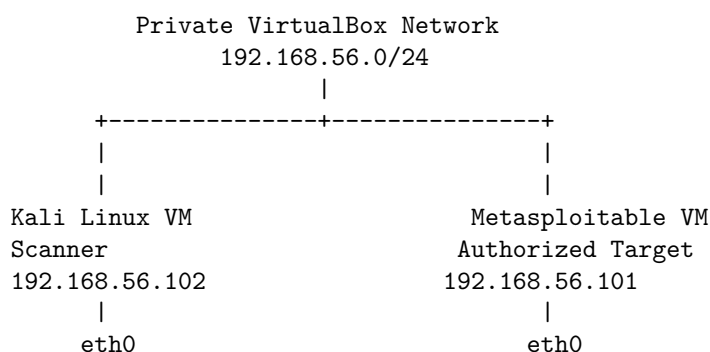
Network

Both systems were located on the same IPv4 subnet:

192.168.56.0/24

The Kali routing table confirmed that the local subnet was directly reachable through eth0.

5. Network Topology



The private lab network provided an isolated environment for performing the assessment.

6. Kali Linux Network Configuration

The Kali Linux interface configuration was collected using:

```
ip a
```

The relevant configuration was:

```
Interface: eth0
IPv4: 192.168.56.102/24
Broadcast: 192.168.56.255
```

The routing table was collected using:

```
ip route
```

The resulting route confirmed:

```
192.168.56.0/24 dev eth0 proto kernel scope link src 192.168.56.102
```

This confirmed that the target subnet was directly connected to the Kali eth0 interface.

Evidence

Figure 1 — Kali Linux IP configuration.

7. Metasploitable Target Identification

The Metasploitable machine was inspected using `ifconfig`.

The target was identified with:

```
IPv4 Address: 192.168.56.101
Subnet Mask: 255.255.255.0
```

Therefore, the authorized target selected for the assessment was:

```
192.168.56.101
```

Evidence

Figure 2 — Metasploitable network configuration and target IP identification.

```

(yuvaraj@kali) ~/Trios-Cyber-Internship
$ cd Day-02-Local-Target-Nmap
(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ pwd
/home/yuvaraj/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ find -maxdepth 2 -type d | sort
./logs
./screenshots
(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:53:7f:c9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 546sec preferred_lft 546sec
    inet6 fe80::a00:27ff:fe53:7fc9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ ip route
192.168.56.0/24 dev eth0 proto kernel scope link src 192.168.56.102 metric 100
(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ ip a | tee logs/kali-ip.txt
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:53:7f:c9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 522sec preferred_lft 522sec
    inet6 fe80::a00:27ff:fe53:7fc9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ ip route | tee logs/kali-route.txt
192.168.56.0/24 dev eth0 proto kernel scope link src 192.168.56.102 metric 100

```

Figure 1: Kali IP configuration

```

collisions:0 txqueuelen:1000
RX bytes:1188 (1.1 KB) TX bytes:3924 (3.8 KB)
Base address:0xd020 Memory:f0200000-f0220000

lo: Link encap:Local Loopback
    inet addr:127.0.0.1 Mask:255.0.0.0
    inet6 addr: ::1/128 Scope:Host
    UP LOOPBACK RUNNING MTU:16436 Metric:1
    RX packets:101 errors:0 dropped:0 overruns:0 frame:0
    TX packets:101 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:0
    RX bytes:23573 (23.0 KB) TX bytes:23573 (23.0 KB)

msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:48:45:d9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe48:45d9/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$

```

Figure 2: Metasploitable IP configuration

8. Connectivity Verification

Connectivity between Kali Linux and the Metasploitable target was verified using ICMP.

Command executed:

```
ping -c 4 192.168.56.101
```

Results

Metric	Result
Packets transmitted	4
Packets received	4
Packet loss	0%
Minimum RTT	0.900 ms
Average RTT	1.124 ms
Maximum RTT	1.226 ms

The successful ICMP responses confirmed that the target was reachable from Kali.

Neighbour discovery was also performed:

```
ip neigh show 192.168.56.101
```

The target MAC address was resolved as:

```
08:00:27:48:45:d9
```

This confirmed successful Layer 2 neighbour resolution between the two virtual machines.

Evidence

```

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ ping -c 4 192.168.56.75 | tee logs/ping-target.txt
PING 192.168.56.75 (192.168.56.75) 56(84) bytes of data.
From 192.168.56.102 icmp_seq=1 Destination Host Unreachable
From 192.168.56.102 icmp_seq=2 Destination Host Unreachable
From 192.168.56.102 icmp_seq=3 Destination Host Unreachable
From 192.168.56.102 icmp_seq=4 Destination Host Unreachable
--- 192.168.56.75 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3063ms
pipe 3

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ ip neigh show 192.168.56.75 | tee logs/target-neighbor.txt
192.168.56.75 dev eth0 FAILED

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ ping -c 4 192.168.56.101 | tee logs/ping-target.txt
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=2.33 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.68 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=1.90 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=2.15 ms
--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3099ms
rtt min/avg/max/mdev = 1.680/2.014/2.334/0.247 ms

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ ip neigh show 192.168.56.101 | tee logs/target-neighbor.txt
192.168.56.101 dev eth0 lladdr 08:00:27:48:45:d9 REACHABLE

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ ping -c 4 192.168.56.101 | tee logs/ping-target.txt
$ ip neigh show 192.168.56.101 | tee logs/target-neighbor.txt
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=1.17 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.900 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=1.23 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.21 ms
--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3045ms
rtt min/avg/max/mdev = 0.900/1.124/1.226/0.131 ms
192.168.56.101 dev eth0 lladdr 08:00:27:48:45:d9 DELAY

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$

```

Figure 3: Connectivity verification

Figure 3 — Successful connectivity and neighbour verification.

9. Nmap Host Discovery

Nmap host discovery was performed using:

```
nmap -sn 192.168.56.101
```

The `-sn` option performs host discovery without performing a conventional port scan.

Results

Nmap scan report for 192.168.56.101

Host is up (0.0016s latency).

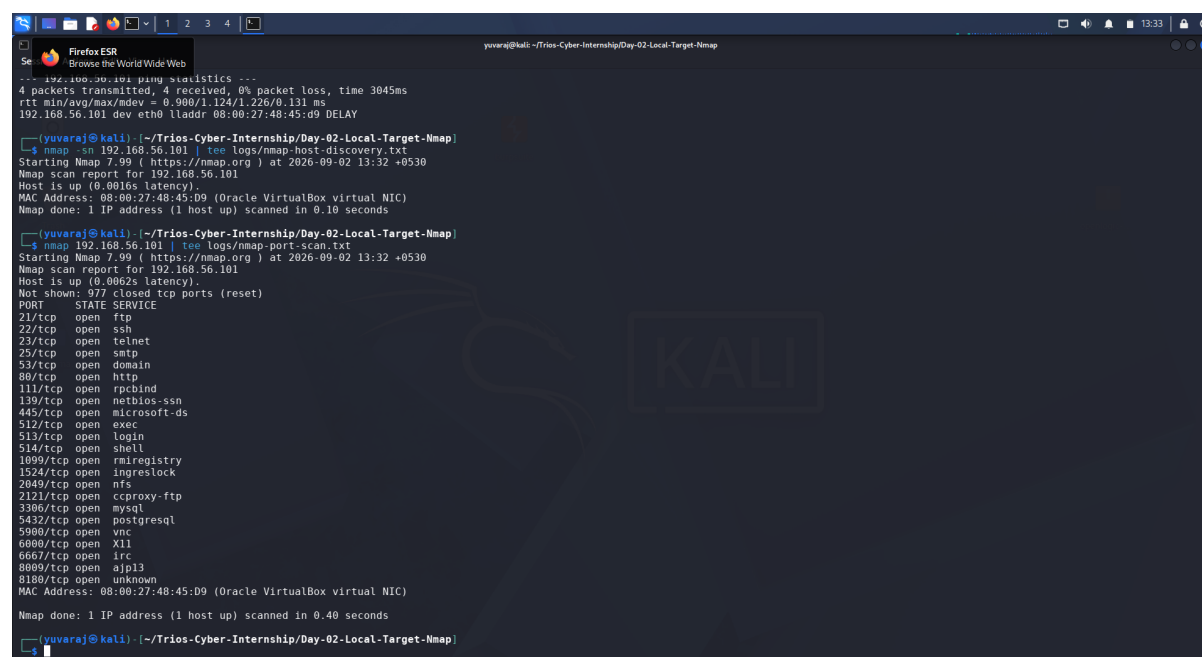
MAC Address: 08:00:27:48:45:D9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds

Findings

- Target host was successfully discovered.
- Target status: **Up**
- Reported latency: **0.0016 seconds**
- MAC address: 08:00:27:48:45:D9
- MAC vendor: Oracle VirtualBox virtual NIC
- Hosts scanned: 1
- Hosts identified as active: 1

Evidence



```
192.168.56.101 ping statistics --
4 packets transmitted, 4 received, 0% packet loss, time 3045ms
rtt min/avg/max/mdev = 0.900/1.124/1.226/0.131 ms
192.168.56.101 dev eth0 lladdr 08:00:27:48:45:D9 DELAY

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ nmap -sn 192.168.56.101 | tee logs/nmap-host-discovery.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-02 13:32 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0016s latency).
MAC Address: 08:00:27:48:45:D9 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ nmap 192.168.56.101 | tee logs/nmap-port-scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-02 13:32 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0025s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rairegistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6080/tcp  open  X11
6881/tcp  open  irc
8080/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:48:45:D9 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.40 seconds

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$
```

Figure 4: Nmap host discovery and port scan

Figure 4 — Nmap host discovery and basic TCP port scan.

10. Basic TCP Port Scan

A basic Nmap TCP scan was performed using:

```
nmap 192.168.56.101
```

The scan identified:

- **23 open TCP ports**
- **977 closed TCP ports**

Discovered Open Ports

Port	State	Nmap Service
21/tcp	open	ftp
22/tcp	open	ssh
23/tcp	open	telnet
25/tcp	open	smtp
53/tcp	open	domain
80/tcp	open	http
111/tcp	open	rpcbind
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds
512/tcp	open	exec
513/tcp	open	login
514/tcp	open	shell
1099/tcp	open	rmiregistry
1524/tcp	open	ingreslock
2049/tcp	open	nfs
2121/tcp	open	ccproxy-ftp
3306/tcp	open	mysql
5432/tcp	open	postgresql
5900/tcp	open	vnc
6000/tcp	open	X11
6667/tcp	open	irc
8009/tcp	open	ajp13
8180/tcp	open	unknown

The result demonstrates that the Metasploitable system exposes a large number of network-accessible services.

11. Service and Version Detection

Nmap service and version detection was performed using:

```
nmap -sV 192.168.56.101
```

The `-sV` option was used to obtain additional information about the services running on the discovered open ports.

Service Detection Results

Port	Service	Detected Version / Information
21/tcp	FTP	vsftpd 2.3.4
22/tcp	SSH	OpenSSH 4.7p1 Debian 8ubuntu1
23/tcp	Telnet	Linux telnetd
25/tcp	SMTP	Postfix smtpd
53/tcp	DNS	ISC BIND 9.4.2
80/tcp	HTTP	Apache httpd 2.2.8 (Ubuntu) DAV/2
111/tcp	RPC	rpcbind 2

Port	Service	Detected Version / Information
139/tcp	NetBIOS	Samba smbd 3.X-4.X
445/tcp	SMB	Samba smbd 3.X-4.X
512/tcp	rexec	netkit-rsh rexecd
513/tcp	login	OpenBSD or Solaris rlogind
514/tcp	shell	Netkit rshd
1099/tcp	Java RMI	GNU Classpath grmiregistry
1524/tcp	Bindshell	Metasploitable root shell
2049/tcp	NFS	NFS 2-4
2121/tcp	FTP	ProFTPD 1.3.1
3306/tcp	MySQL	MySQL 5.0.51a-3ubuntu5
5432/tcp	PostgreSQL	PostgreSQL 8.3.0-8.3.7
5900/tcp	VNC	VNC protocol 3.3
6000/tcp	X11	Access denied
6667/tcp	IRC	UnrealIRCd
8009/tcp	AJP13	Apache Jserv Protocol v1.3
8180/tcp	HTTP	Apache Tomcat/Coyote JSP engine 1.1

Nmap also reported Unix/Linux operating system information and the hostnames:

```
metasploitable.localdomain
irc.Metasploitable.LAN
```

Evidence

```

Session Actions Edit View Help
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:48:45:D9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$ nmap -sV 192.168.56.101 | tee logs/nmap-service-scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-02 13:34 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0049s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          Netkit rshd
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2.4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:48:45:D9 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.26 seconds

(yuvaraj@kali) ~/Trios-Cyber-Internship/Day-02-Local-Target-Nmap
$

```

Figure 5: Nmap service detection

Figure 5 — Nmap service and version detection results.

12. Security Observations

The enumeration results demonstrate that the target exposes a broad range of network services.

The following observations were recorded based strictly on the Nmap results:

12.1 Legacy Remote Access Services

Telnet and rsh-related services were identified:

```
23/tcp  telnet
512/tcp  exec
513/tcp  login
514/tcp  shell
```

These services represent legacy remote-access functionality and increase the network attack surface.

12.2 Multiple File Transfer Services

Two FTP services were identified:

```
21/tcp  vsftpd 2.3.4
2121/tcp ProFTPD 1.3.1
```

The presence of multiple FTP services increases the number of network-accessible application endpoints.

12.3 Network File and Windows Interoperability Services

The target exposed:

```
139/tcp  Samba
445/tcp  Samba
2049/tcp  NFS
```

These services provide network file-sharing functionality and should be carefully controlled in production environments.

12.4 Database Services

Database services were directly accessible:

```
3306/tcp  MySQL
5432/tcp  PostgreSQL
```

Database services generally require strict network access controls and should not be unnecessarily exposed to untrusted networks.

12.5 Remote Display Services

The scan identified:

```
5900/tcp  VNC
6000/tcp  X11
```

These services provide remote graphical access functionality and contribute to the exposed attack surface.

12.6 Web and Application Services

Several web/application-related services were identified:

```
80/tcp    Apache HTTP
8009/tcp  Apache Jserv/AJP13
8180/tcp  Apache Tomcat
```

These services would warrant further security assessment in a dedicated authorized vulnerability-assessment phase.

12.7 Metasploitable Bindshell

Nmap identified:

```
1524/tcp open bindshell Metasploitable root shell
```

This is an especially notable observation because the service identification itself indicates a root-shell service exposed by the intentionally vulnerable training target.

No connection or exploitation of this service was performed during this assessment.

13. Assessment Methodology

The assessment followed a controlled enumeration workflow:

Step 1 — Network Configuration

The Kali Linux network interface and routing information were inspected using:

```
ip a  
ip route
```

Step 2 — Target Identification

The Metasploitable system was inspected locally to identify its assigned IP address.

Step 3 — Connectivity Verification

Connectivity was tested using:

```
ping -c 4 192.168.56.101
```

Neighbour resolution was verified using:

```
ip neigh show 192.168.56.101
```

Step 4 — Host Discovery

Nmap host discovery was performed:

```
nmap -sn 192.168.56.101
```

Step 5 — Basic Port Enumeration

A standard TCP scan was performed:

```
nmap 192.168.56.101
```

Step 6 — Service Detection

Service and version identification was performed:

```
nmap -sV 192.168.56.101
```

Step 7 — Evidence Preservation

Command outputs were saved using `tee` into the `logs/` directory, while screenshots were retained in the `screenshots/` directory.

14. Evidence Index

Evidence ID	Description	File
E01	Kali IP configuration	screenshots/01-kali-ip.png
E02	Metasploitable IP configuration	screenshots/02-metasploitable-ip.png
E03	Connectivity verification	screenshots/03-connectivity-verification.
E04	Nmap host discovery and basic port scan	screenshots/04-nmap-discovery-and-port-sc
E05	Nmap service/version detection	screenshots/05-nmap-service-detection.png

Raw Command Logs

Log	Purpose
logs/kali-ip.txt	Kali interface configuration
logs/kali-route.txt	Kali routing information
logs/ping-target.txt	ICMP connectivity results
logs/target-neighbor.txt	Target neighbour/MAC resolution
logs/nmap-host-discovery.txt	Nmap host discovery results
logs/nmap-port-scan.txt	Basic TCP port scan results
logs/nmap-service-scan.txt	Service/version detection results

15. Key Findings Summary

Category	Result
Kali IPv4	192.168.56.102/24
Target IPv4	192.168.56.101/24
Network	192.168.56.0/24
Connectivity	Successful
Packet Loss	0%
Target Host	Up
Open TCP Ports	23
Closed TCP Ports	977
Service Detection	Successful
Target OS Family	Unix/Linux
Exploitation Performed	No

16. Conclusion

The Day 02 assessment successfully established and validated an authorized local target lab using Kali Linux and Metasploitable in a private VirtualBox network.

Connectivity between the assessment machine and target was successfully verified, and Nmap confirmed that the Metasploitable host was active.

The basic TCP scan identified 23 open ports, while service/version detection provided detailed information about the applications exposed by the target. The assessment demonstrated practical use of Nmap for host discovery, port enumeration, and service identification.

The results also demonstrated how a deliberately vulnerable training system can expose a large network attack surface through legacy remote-access services, file-sharing protocols, databases, web applications, and remote graphical services.

No exploitation or unauthorized activity was performed. The assessment remained within the defined local laboratory scope.

This assessment establishes the foundation for future authorized security-testing activities by demonstrating the initial reconnaissance and enumeration stage of a penetration-testing workflow.

17. Appendix — Commands Executed

Kali Network Identification

```
ip a
ip route
```

Target Connectivity

```
ping -c 4 192.168.56.101
ip neigh show 192.168.56.101
```

Nmap Host Discovery

```
nmap -sn 192.168.56.101
```

Basic TCP Port Scan

```
nmap 192.168.56.101
```

Service and Version Detection

```
nmap -sV 192.168.56.101
```

Evidence Storage

```
tee logs/<filename>.txt
```

All commands were executed against the authorized local Metasploitable target at 192.168.56.101.

End of Day 02 Assessment Report